

# **Companion to Bird & de Moor, Algebra of Programming**

## §1. Programs

### §1.1. Datatypes

## §1.2. Natural numbers

## §1.3. Lists

## §1.4. Trees

## §1.5. Inverses

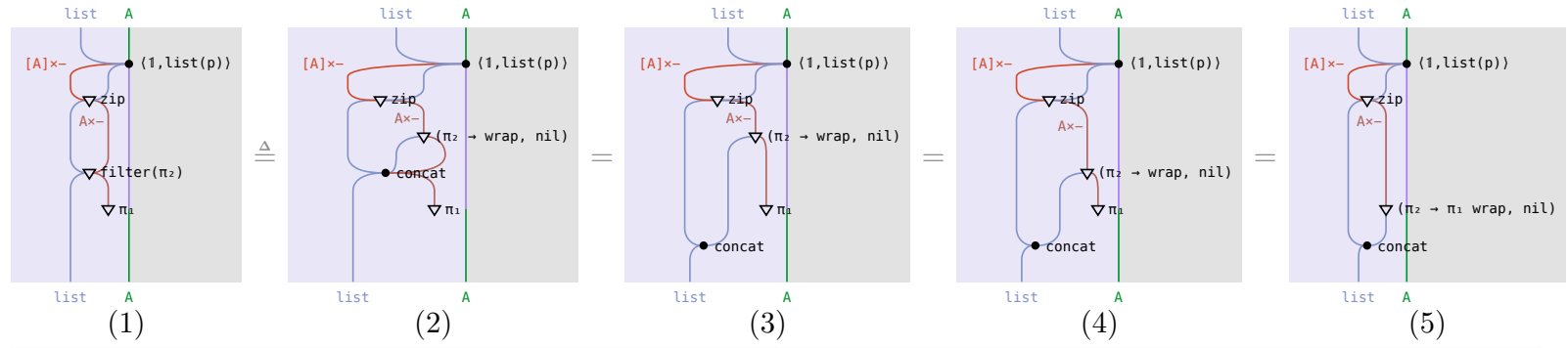
## §1.6. Polymorphic functions

## §1.7. Pointwise and point-free

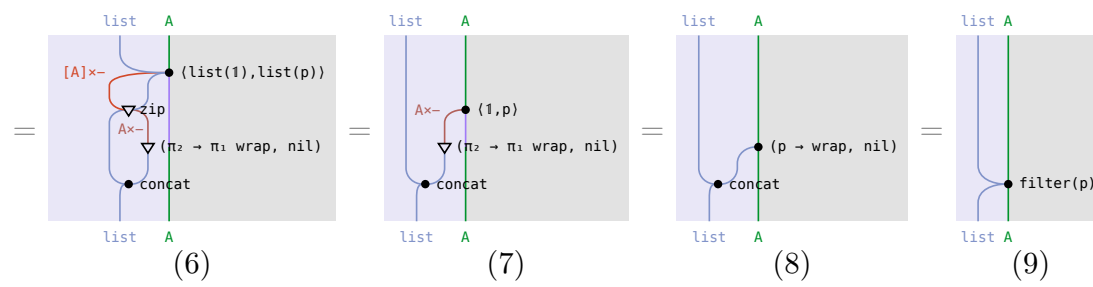
$$\langle 1, \text{list}(p) \rangle \text{ zip filter}(\pi_2) \text{ list}(\pi_1) = \text{filter}(p) \quad (1.7a)$$

pairing each element with its answer under  $p$  ( $\langle 1, \text{list}(p) \rangle \text{ zip}$ ), keeping the pairs answered **true** ( $\text{filter}(\pi_2)$ ) and dropping the answers ( $\text{list}(\pi_1)$ ) is  $\text{filter}(p)$

the starting composite



|   |                                                                                                                                                     |                                                                                                                                                     |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\langle 1, \text{list}(p) \rangle \text{ zip filter}(\pi_2) \text{ list}(\pi_1)$                                                                   | definition of filter: $\text{filter}(p) \triangleq \text{list}((p \rightarrow \text{wrap}, \text{nil}))$<br>concat                                  |
| 2 | $\triangleq \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \text{wrap}, \text{nil})) \text{ concat list}(\pi_1)$             | (1.6): $\text{concat list}(f) = \text{list}(\text{list}(f)) \text{ concat}$                                                                         |
| 3 | $= \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \text{wrap}, \text{nil})) \text{ list}(\text{list}(\pi_1)) \text{ concat}$ | (1.8) backwards: $\text{list}(f) \text{ list}(g) = \text{list}(fg)$                                                                                 |
| 4 | $= \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \text{wrap}, \text{nil}) \text{ list}(\pi_1)) \text{ concat}$              | (1.11): $(p \rightarrow f, g)h = (p \rightarrow fh, gh)$ ; (1.5): $\text{wrap list}(f) = f \text{ wrap}$ ; (1.4): $\text{nil list}(f) = \text{nil}$ |
| 5 | $= \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})) \text{ concat}$                           | (1.9) backwards: $1 = \text{list}(1)$                                                                                                               |



|   |                                                                                                                                        |                                                                                                                                                                                |
|---|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | $= \langle \text{list}(1), \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})) \text{ concat}$ | (1.7): $\langle \text{list}(f), \text{list}(g) \rangle \text{ zip} = \text{list}(\langle f, g \rangle)$<br>(1.8) backwards: $\text{list}(f) \text{ list}(g) = \text{list}(fg)$ |
| 7 | $= \text{list}(\langle 1, p \rangle (\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})) \text{ concat}$                                | (1.10): $h(p \rightarrow f, g) = (hp \rightarrow hf, hg)$ ; (1.1): $\langle f, g \rangle \pi_1 = f$ ; (1.3): $f \text{ nil} = \text{nil}$                                      |
| 8 | $= \text{list}((p \rightarrow \text{wrap}, \text{nil})) \text{ concat}$                                                                | (1.12): $1 f = f$                                                                                                                                                              |
| 9 | $= \text{filter}(p)$                                                                                                                   | definition of filter                                                                                                                                                           |